

SECTION B

Answer all questions in the spaces provided.

8. (a) Sodium chloride and sodium bromide both react with concentrated sulfuric acid. Describe the observations in both reactions and explain why they are different. [3]

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- (b) Fluorine can form the weak acid hydrofluoric acid, HF. This acid has a K_a value of $7.20 \times 10^{-4} \text{ mol dm}^{-3}$.

- (i) Write an expression for the K_a of hydrofluoric acid. [1]

- (ii) Calculate the pH of a solution of hydrofluoric acid of concentration $0.100 \text{ mol dm}^{-3}$. [3]

pH =



- (iii) The concentration of a solution of hydrofluoric acid can be found by titrating against sodium hydroxide. Not all acid-base indicators would be suitable for this titration. Explain what features would make an indicator suitable for use in a weak acid-strong base titration. [2]

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- (iv) Addition of 12.5 cm^3 of 0.100 mol dm^{-3} sodium hydroxide to 25.0 cm^3 of hydrofluoric acid of concentration 0.100 mol dm^{-3} forms a buffer solution.

- I. Explain how this buffer solution works. [3]

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- II. Calculate the pH of this buffer solution. [2]

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